Paleomagnetism of the Mealy Mountain Anorthosite Suite and of the Shabogamo Gabbro, Labrador, Canada

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Two stable, ancient, N.R.M. components were isolated by a.f. demagnetization of samples from 29 sites in the Helikian (Middle Proterozoic) anorthosite and adamellite suites of the Mealy Mountains. The directions of the two components are chiefly to the northwest, 20° upward (northwest component) and to the east, 65° downward (east component). The northwest direction is thought to be the primary igneous T.R.M. and no reversely magnetized sites with this direction were encountered in the Mealy suite. The 11 sites from which the northwesterly component was isolated yielded a mean pole at 179° W, 8° N, $\alpha_{95} = 12°$. This is some distance away from poles derived from rocks of comparable age from elsewhere in the Canadian Shield and may reflect a 50° clockwise rotation of this section of the Grenville Orogen relative to the Superior after the initial magnetization of the Mealy rocks. Evidence of a second stable component was found in 14 sites (east component); 11 of these are reversely magnetized and three are normally magnetized. The 14 sites with easterly (and reversed easterly) magnetization yielded a combined pole for this component at 182° W, 38° S, $\alpha_{95} = 9°$. This pole position suggests that the east component was acquired after the rotation of this part of the Grenville Orogen. The cores from 3 sites exhibited two clear end-points during a.f. demagnetization and thus contain both the northwest and the east component.

The Helikian Shabogamo Gabbro at the south end of the Labrador Trough was sampled at 13 sites on a line extending 80 km north from the Grenville Front. After a.f. cleaning these sites yielded a pole at 171° W, 10° N, $\alpha_{95} = 12°$. This pole lies about 20° northwest of poles for rocks of similar age in the Superior Orogen so it is suggested that the rocks of the sample area, though lying a bit north of the line generally accepted as the 'Grenville Front', have been rotated clockwise (about 20°) since intrusion of the gabbro.

Deux composantes stables anciennes N.R.M. ont été reconnues par démagnétisation a.f. d'échantillons provenant de 29 localités situées dans des séquences d'anorthosite et d'adamellite des Monts Mealy. Les directions de ces deux composantes sont principalement nord-ouest, 20° vers le haut (composante nord-ouest) et est, 65° vers le bas (composante est). Nous pensons que la direction nord-ouest est la T.R.M. primaire igné; aucune autre localité de magnétisation inverse avec cette direction n'a été reconnue dans la séquence Mealy. Les onzes localités qui ont permis d'isoler la composante nord-ouest donnent un pôle moyen à 179° W, 8° N, $\alpha_{95} = 12°$. Celui-ci diffère appréciablement des pôles dérivés de roches d'âge semblable venant d'ailleurs dans le Bouclier Canadien et pourrait signifier qu'il y a eu une rotation de 50°, sens horaire, de ce secteur de l'orogène grenvillien par rapport à celui du Supérieur, après la magnétisation des roches de la séquence Mealy. L'évidence de la deuxième composante stable (direction est) provient de 14 localités dont 11 montrent une magnétisation inversée et trois une magnétisation normale. Ces 14 localités de magnétisation est (normale et inversée) furnissent un pôle moyen de 182° W, 38° S, $\alpha_{95} = 9°$. Cette position laisse entendre que la composante est a été acquise après la rotation de ce secteur de l'orogène grenvillien. Les carottes de trois localités ont montré deux points terminaux libres pendant la démagnétisation a.f. et ainsi contiennent les deux composantes nord-ouest et est.

Le Gabbro hélikien de Shabogamo à l'extrémité sud de la Fosse du Labrador a été échantillonné à 13 endroits dans une ligne que se prolonge jusqu'à 80 km au nord du Front grenvillien. Après un lavage a.f. les échantillons ont donné un pôle à 171° W, 10° N, $\alpha_{95} = 12°$. Ce pôle est à environ 20° au nord-ouest des pôles de roches d'âge semblables de l'orogène Supérieur ce qui pourrait signifier que les roches de cette région quoique situées un peu au nord de la ligne généralement acceptée comme celle du 'Front grenvillien' ont subi une rotation sens horaire (environ 20°) depuis l'intrusion du gabbro.

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Geology of the Mealy Mountain Anorthosite

The anorthosites of the Mealy Mountains, and those rocks to the south and southwest that appear to be genetically related to the anorthosites, have been mapped only on a reconnaissance scale; the area east of Longitude 60° at 1 in. = 8 mi (Eade 1962); the area west of this at 1 in. = 4 mi (Stevenson 1967*a*, *b*) (Fig. 1). The Mealy Mountains consist essentially of relatively fresh, locally layered, dark gray to black anorthosite. This mass is terminated on the northeast by a major

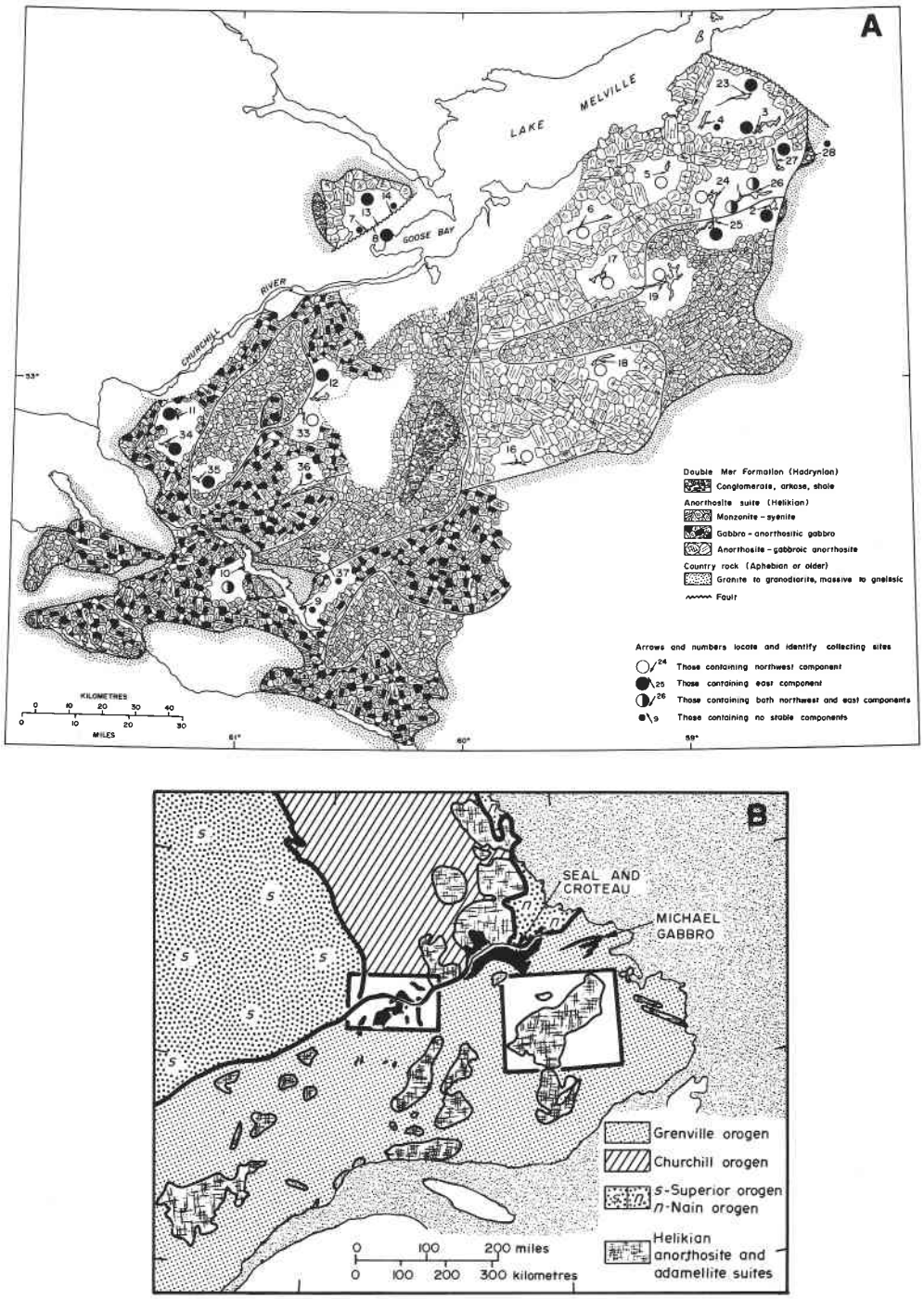


FIG. 1.(A) Distribution of anorthosite and associated monzonites and gabbros in the Mealy Mountains region (from Eade 1962 and Stevenson 1967*a*, *b*). (B) Index map showing location and regional geological relationships of the Mealy Mountain and Shabogamo rocks. The two rectangles outline areas shown in greater detail in Figs. 1(A) and 6.

fault and grades southward and westward into a syenite-monzonite series of rocks. These have a distinctive green resinous fresh surface and typically exhibit a deeply-weathered, rusty-brown outcrop surface. The monzonitic rocks give way farther west to predominantly gabbroic associates. Both Eade and Stevenson have emphasized the gradational nature of contacts between all of these rock types and Stevenson has considered them to be a differentiated series.

No radioisotopic dating has yet been carried out on the Mealy Mountain rocks, however it is likely that they are similar in age to the Harp Lake and Michikamau intrusions to the northwest. Age determinations have been carried out on rocks whose relationship to the Harp and Michikamau intrusions provide some insight into their ages. K-Ar ages on rocks near the Michikamau (Emslie *in* Wanless *et al.* 1965) suggested an age of about 1500 m.y. for the Michikamau. The Harp Lake intrusion is unconformably overlain by strata of the Seal Group whose igneous components have a magnetization direction similar to those of the Croteau Group (Roy and Fahrig 1973). The Croteau has yielded a Rb/Sr age of 1562 ± 42 m.y.[1] (Wanless and Loveridge 1972) so the Harp Lake intrusion is probably in the range 1500–1600 m.y. old. Morse (1964) has reported a 1480 ± 50 m.y. K-Ar age on biotite from possibly rheomorphic granodiorite that cuts the Kiglapait anorthosite; an intrusion which lies farther north on the Labrador Coast. As the K-Ar ages from this rock and from those near Michikamau are probably minimum ages the data are consistent with an age of 1500–1600 m.y. for all three intrusions and it is quite reasonable to assume that the Mealy Mountain anorthosite had an original age of crystallization in this same range.

Paleomagnetism of the Mealy Mountain Anorthosite

A total of 191 oriented drill cores, 2.5 cm in diameter, were collected from the 29 sites whose location is indicated by Fig. 1. The cores were oriented by sun compass and measured with the Geological Survey of Canada automated biastatic magnetometer (Larochelle and Christie 1967). Two specimens were cut from each core and the mean value for the two specimens was used as basic datum for those cores that were homogeneously magnetized (θ, the angle between two specimens of the same core, is less than 25°) after appropriate partial alternating field demagnetization.

At least two specimens from each site were step-cleaned in fields from 50 to 1000 Oe in order to choose the optimum demagnetizing field for each site. The test specimens for most sites exhibited one, or less commonly, two end-points during the step-cleaning procedure. All of the specimens from sites exhibiting one or more end-points were then cleaned in at least three field intensities in the range suggested by step-cleaning.

It was found impossible to isolate and measure a stable remanent magnetization direction in samples from 7 of the 29 sites. This is thought to be due to a variety of reasons. The rock at sites 7 and 14 is close to a fault and has been metamorphosed to form a light gray granular aggregate whose magnetization is weak and unstable. In the other sites, the step-cleaned specimens exhibited no end-points and step-cleaning the remaining specimens from these sites did not produce an improved grouping. Many of these sites are from gabbroic or monzonitic rocks. It is likely that in some sites the coercivity spectra of two stable components overlap to a large extent so that the two components cannot be separated with a.f. demagnetization. It is also likely that some of the sites do not contain a stable remanent magnetization. The magnetization of gabbro from the Kiglapait intrusion near the Labrador coast was found to be unstable when these were examined in a pilot project several years ago. Instability may be a characteristic common to gabbros related to anorthosite intrusions.

Stable magnetization was isolated in 22 of the 29 Mealy Mountain sites. This stable magnetization consists of two components; one of normal polarity (*i.e.* it yields a North pole in the Pacific) and pointing up in the northwest quadrant (northwest component); the other predominantly of reversed polarity and steeply down to the east (east component), but with 3 sites of normal polarity (*i.e.* steeply up to the west). These 3 sites were grouped with the east component because when reversed they are

[1]Decay, constant $1.39 \times 10^{-11} \times yr^{-1}$.

statistically closer to the east component direction than to the northwest component. Eight sites exhibited only the northwest component, eleven only the east component and three sites both the northwest and east components. It is emphasized that the sites yielding a northwest direction and those yielding a predominantly east direction are of normal and predominantly reversed polarity respectively, but they are far from being exactly reversed relative to one another.

The relationship of the two components can be well illustrated by the behavior during step cleaning of cores from site 26, which exhibits both components (Fig. 2). Cleaning at 100 Oe has removed a soft component (presumably related to the recent earth's field direction) and 6 of the 7 cores are well grouped. Core 2 however, has begun to shift direction, and the angle (θ) between the two specimens from this core has increased considerably. This is quite characteristic behavior as the cores gradually lose their strong but less stable northwest component and probably indicates that individual cores are not internally homogeneous with respect to the two components. In any case, the cores whose magnetization direction reaches a stable end-point in a steep east direction generally exhibit a small θ after reaching this end-point direction. After cleaning at 500 Oe, 3 cores still exhibit only the northwest components but the other 4 cores exhibit the east component (although core 6 is still a bit outside the group and has a high θ). After cleaning at 800 Oe, and higher, the directions are essentially random. Two distinct components were isolated in rocks from this site so it was treated statistically as a double site, as were sites 1 and 10, Table 1. The northwest component for site 26 was defined by data from 6 cores after cleaning at 100 Oe and the east component by 3 cores after cleaning at 500 Oe.

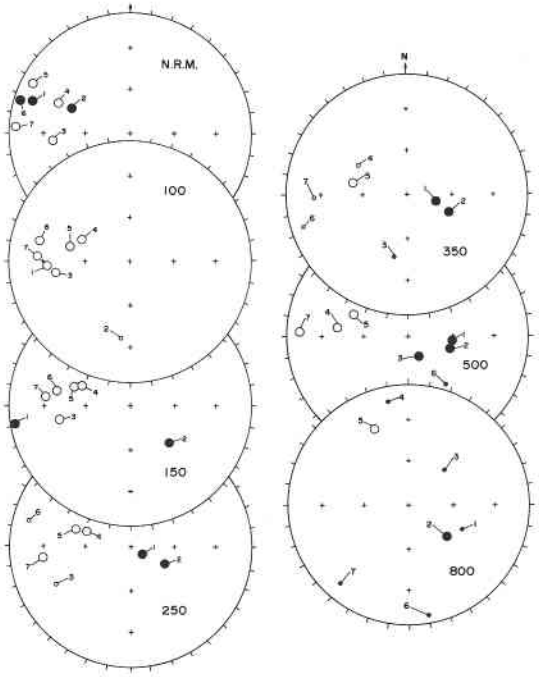


FIG. 2. Magnetization directions of cores from site 26 during step a.f. cleaning. Solid circles, north-seeking down, open circles north-seeking up. Small circles indicate that θ (angle between two specimens from the same core) exceeded 25°. Equal area projection.

Discussion of Mealy Mountain Paleomagnetic Results

The within-site and between-site grouping of directions of both the northwest and east components showed general improvement as a result of a.f. partial demagnetization (Table 1, Fig. 3). This improvement was notably greater for the east component than it was for the northwest component, and the final grouping after optimum cleaning was much tighter than that of the northwest component. Both components have optimum cleaning fields with a wide range and the mean optimum a.f. cleaning fields are similar. The step-cleaning of those cores from sites 1 and 26 that clearly contain both components suggests that in these cores at least, the coercivity of the east component (as indicated by the intensity of the destructive alternating field) is higher than that for the northwest component. This may mean that random soft components have not in every case been as successfully removed by a.f. cleaning from cores dominated by the northwest component. However, magnetic directions of cores exhibiting the northwest component are not streaked towards the east component direction. It is unlikely then that failure to remove an east component is a major factor in the poorer grouping of the northwest component. This may have been a factor in some sites which were eliminated because they failed to exhibit an end-point during a.f. cleaning.

TABLE 1. Directions of remanent magnetization before (NRM), and direction of magnetization and pole positions after magnetic cleaning

			Mealy Mountains–northwest component NRM									
Site	N	n	D	I	$\delta°$	k	a.f.	D	I	$\delta°$	k	δ°_{R}
1	7	6	266	4	43	4	100	271	−49	8.5	90	3.5
5	7	3	296	−21	40	4	800	320	−35	13	37	7.7
6	6	5	291	− 9	25	11	150	290	−10	15	28	6.9
10	7	5	280	47	46	3	200	310	24	9.4	74	4.2
16	7	3	331	44	57	2	350	318	−31	5.9	187	3.4
17	6	6	290	8	8	98	500	290	− 2	4.9	277	2.0
18	7	7	326	47	11	56	200	323	−15	4.0	418	1.5
19	6	5	316	−33	28	8	800	302	−38	9.3	76	4.1
24	5	5	282	−34	22	14	800	287	−37	8.5	91	3.8
26	7	6	284	−11	24	17	100	276	−34	15	28	6.2
33	6	5	313	66	12	46	350	302	37	13	36	6.0
All cores 56			294	10	44	3		298	−18	31	7	4.2
All sites 11			295	10	39	4		300	−19	31	7	9.3
Mean (north) pole position from 11 cleaned sites listed above (northwest component)—							179.3 °W		8.4 °N	$k = 12$	$\alpha°_{95} = 12.0°$	
			Mealy Mountains–east component									
1	7	2*	296	53	10	63	400	032	69	13	39	9.2
2N'	7	7	063	−46	70	1	150	090	51	4.1	394	1.5
3N'	9	9	093	62	25	11	500	084	70	5.6	206	1.9
8	5	2	127	− 2	48	3	150	088	49	21	15	15
10	7	2*	226	67	42	4	150	116	81	12	47	8.4
11	7	7	113	14	58	2	500	092	62	4.5	330	1.7
12	7	7	051	44	34	6	500	068	66	5.4	228	2.0
13	6	3	030	28	34	6	500	061	68	11	54	1.4
23	8	6	097	74	8	101	350	131	67	9.4	75	3.8
25N'	6	4	118	52	20	17	250	104	58	8.4	97	4.2
26	7	3*	282	11	43	4	500	112	66	14	36	7.8
27	6	4	011	79	16	25	200	102	56	12	44	6.1
34	6	6	063	69	9	82	500	070	67	3.4	572	1.4
35	6	4	021	70	20	17	350	072	50	4.8	288	2.4
All cores 66			073	65	49	3		088	64	14	35	1.7
All sites 14			069	69	49	3		087	65	14	35	3.6
Mean (south) pole position from 14 cleaned sites listed above (east component)—							181.7 °W		37.9 °S	$k = 17$	$\alpha°_{95} = 9.1$	

NOTES: N = number of cores collected at the site; n = number of samples left after excluding those cores that are inhomogeneously magnetized (θ greater than 25°). Numbers under the n column that are marked with an asterisk are special cases, which are explained in the text. D = declination, I = inclination, $\delta°$ = angular standard deviation, k = Fisher's estimate of precision, a.f. = peak alternating cleaning field in oersteds, δ°_{R} = standard error of the mean, $\alpha°_{95}$ = half angle of the cone of confidence. The magnetization of sites 2, 3 and 25 (marked N') is reversed relative to the remaining sites of this group. The directions from these sites have been reversed in order to mean them with the east component group. Mean pole positions were calculated from site pole positions.

A more probable explanation lies in the clockwise rotation of this part of the Grenville Province after the development of the northwest component. Figure 4 indicates that simple rotation[2] of the Mealy Mountain area about a local axis will bring the pole for the northwest component into good agreement with that of the Michikamau anorthosite (Murthy *et al.* 1968). The northwest component is thus probably a primary T.R.M. and the greater dispersion of this component is attributed to differential movements within the anorthosite during rotation of this crustal segment.

The pole for the east component lies in the South Pacific (near New Zealand) and close to the southern limit of the Grenville Loop

[2]Tilting about a horizontal axis was not considered to be a likely explanation of the pole position because the area involved would be very large, including besides the Mealy Mountain anorthosite, the Seal and Croteau Groups and Michael Gabbro (Fahrig and Larochelle 1972; Roy and Fahrig 1973).

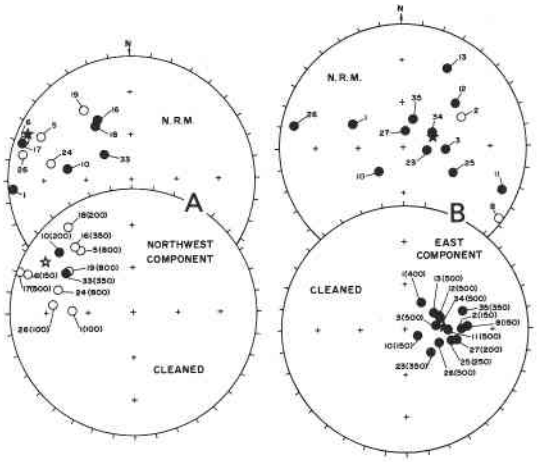


FIG. 3. Site mean directions of Mealy Mountain northwest component before, and after a.f. cleaning (A) all sites of normal polarity. Site mean directions of Mealy Mountain east component before and after a.f. cleaning (B) includes eleven sites of reversed polarity (steeply down to the east) and three sites of normal polarity, which have been reversed for plotting. Sites are identified by numbers and optimum cleaning fields are indicated by bracketed numbers. Stars are group directions giving each site unit weight.

(Roy and Fahrig 1973). This component appears to postdate the suggested clockwise rotation of the Grenville Orogen and the close grouping of this component supports this hypothesis. The steep east (and related reversed) component cannot then be a primary T.R.M., at least in the anorthosites, but is presumably related to thermal events that postdated intrusion and rotation of the anorthosites of the Mealy Mountains. The sites exhibiting this component are distributed around the perimeter of the main anorthosite core, largely in the monzonite–gabbro series. This distribution and position of the pole for this component might suggest that these rocks were intruded much later than the anorthosite, but gradational contacts between these rocks and the anorthosites (Eade 1962; Stevenson 1967*a*, *b*) is an argument against this. The monzonite–gabbro rocks may, because of petrologic differences and areal distribution, have been more susceptible to a regional thermal event that followed the intrusion of the Mealy Mountain anorthosite suite and postdated rotation of this part of the Grenville Orogen. The pole position of the east component relative to the polar wandering curve (Roy and Fahrig 1973) suggests that the rotational event had an age of about 1350 m.y.

Magnetic Mineralogy

The presence of two stable NRM components suggested carrying out a thermomagnetic study on Mealy Mountain samples. Six specimens (one from each of sites 1, 6, 11, 17, 26, 34) were analyzed with a recording automatic magnetic force balance.

The thermomagnetic curves for specimens from the sites with only an east or northwest directed NRM component show the presence of only one magnetic mineral with a Curie point between 550 and 570 °C. A representative curve is given in Figure 5*A*. Apparently, the two different NRM components are not due to a systematic difference in the magnetic mineralogy.

The specimens from sites 1 and 26, containing both components, show the presence of a low Curie point (~300 °C) phase in addition to a high Curie point phase (555–565 °C). In terms of contribution to the total saturation magnetization of the specimens, the low Curie point phase is of minor importance (Fig. 5*B*). The low Curie point phase presumably is a highly titaniferous iron oxide while the high Curie point is assumed to represent virtually pure magnetite. Since those cores having only the northwest component and those having only the east component have identical magnetic mineralogy, it is unlikely that the low Curie temperature phase found in cores from sites 1 and 26 plays a significant role in the remanence directions.

Geology of the Shabogamo Gabbro

The Shabogamo Gabbro comprises an extensive belt of diabasic olivine and anorthositic gabbro intrusions that straddles the Grenville Front at the south end of the Labrador Trough (Fig. 6 and 1*B*). The Gabbros intrude the Sims Formation, which lies unconformably on folded strata of the Kaniapiskau Supergroup. Near the Grenville Front, and to the south, the gabbro has been deformed, metamorphosed, and intruded by granitic and pegmatitic dikes (Eade 1952; Fahrig 1960; 1967; Wynne-Edwards 1960). Parts of the belt south of the Front have undergone intense cataclasis (along with the gneisses that they intrude) and elsewhere, south of the Front, the gabbros are

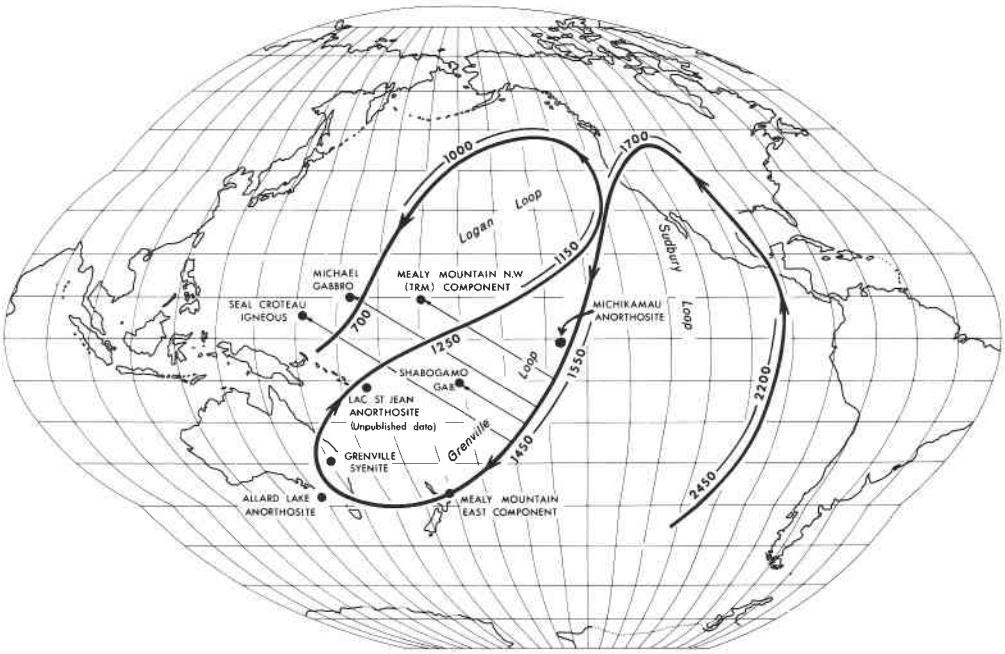


FIG. 4. The polar wandering curve for the Superior Orogen of North America. Logan Loop from Robertson and Fahrig (1972). Grenville Loop, modified from Roy and Fahrig (1973), Sudbury Loop, Fahrig and Chown (1973). The Seal-Croteau, Michael, and Mealy Mountain (northwest component) poles are thought to have moved as indicated due to rotation of the Northeast Grenville Orogen between 1350 and 1550 m.y. ago.

converted to pyroxene plagioclase gneiss. The larger Shabogamo intrusions are probably gently to moderately dipping sheets.

There have been no radioisotopic age determinations carried out on Shabogamo gabbro; the relationships described above, however, indicate that they are Helikian in age. The presence of anorthositic phases in the gabbro may suggest affinities with the major anorthosite intrusions of the Eastern Shield. If this is the case, the Shabogamo Gabbro has an age of about 1500 m.y.

The intrusive and supracrustal rocks of the northern part of the Grenville Orogen, for example, the Michael Gabbro (Fahrig and Larochelle 1972) and the Seal and Croteau Groups (Roy and Fahrig 1973) provide paleomagnetic evidence of having been rotated relative to the older orogens of the Shield. Paleomagnetic study of the Shabogamo Gabbro was carried out to examine the extent of this effect north of the Front. The sites extend about 80 km northeast from the Front as shown by Stockwell (1969).

Paleomagnetism of the Shabogamo Gabbro

The techniques used in isolating and measuring the stable remanent magnetization of the Shabogamo were the same as those described in dealing with the Mealy Mountain gabbro. The test specimens exhibited good end-points and the magnetization directions of the samples moved away from the present earth's field direction, suggesting the elimination of a soft component related to the recent field direction (Table 2, Fig. 7). With the exception of site 50, the within-site grouping either improved in response to a.f. demagnetization or remained unchanged. The between-site grouping, and the grouping of all cores (giving each core unit weight) did not change significantly with cleaning (Table 2). The stability of the magnetization of the Shabogamo as suggested by the optimum a.f. cleaning field is quite high (Table 2), comparable to that of the Mealy Mountain anorthosite (this paper) or the Michael Gabbro (Fahrig and Larochelle 1972). The site mean declinations show greater variation than do the site mean in-

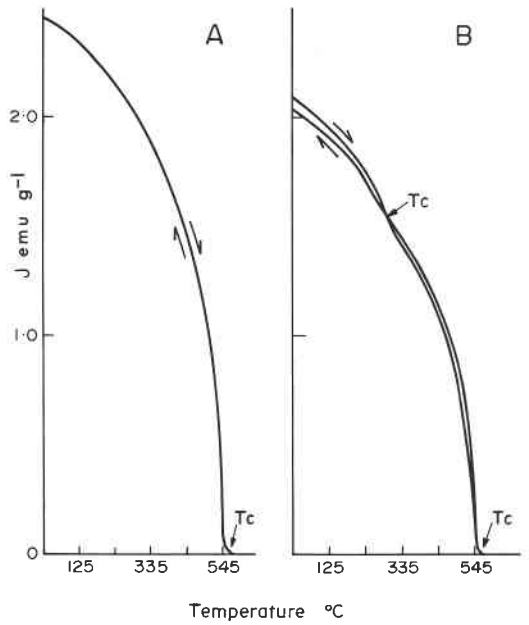


FIG. 5. The change in magnetization J, in e.m.u. g^{-1} during heating and cooling of a specimen from site 34, (A) above left, and site 26, (B) above right, in a constant magnetic field of 5350 Oe. *Tc* is the Curie temperature. Note that in (B) two Curie temperatures are apparent at about 300° and 555–565 °C. Air pressure in the instrument was 10^{-2} torr.

clinations. The magnetization direction of rock at sites 49 to 52, is anticlockwise relative to directions found in 8 of the 9 other Shabogamo sites (Fig. 7). These 4 sites are farthest from the Grenville Front.

The mean pole position of the Shabogamo Gabbro is about 20° northwest of poles for rocks that may be of comparable age, but which lie undisturbed on the Superior Orogen, for example, that of the Sibley Group, (Robertson 1973). If this implies rotation of this area of crust relative to the Superior Orogen, the rotation of the Shabogamo is in the same sense, but less than half that suggested by rocks of comparable age from within the Grenville (Michael Gabbro, Fahrig and Larochelle 1972; Seal-Croteau igneous, Roy and Fahrig 1973; Mealy Mountain northwest component, this paper). This is compatible with the location of most of the Shabogamo sites that lie just north of the Grenville Front; and it is consistent with the apparent lack of rotation suggested by sites 49 to 52, which lie farther north. The Shabogamo between-site grouping ($k = 8$) is similar to that found for the northwest component of the Mealy Mountain anorthosite ($k = 7$) and as in the case of the Mealy Mountain rocks may reflect differential movement within the sample area. The amount of rotation of the Shabogamo rocks apparently decreases northward away from the Front.

Interpretation of Grenville Paleomagnetism

The geology of the Grenville Orogen vis-a-vis the geology of neighboring more ancient orogens imposes certain constraints on the interpretation of Grenville paleomagnetic data. Early Proterozoic strata of the Kaniapiskau Supergroup, for example, can be followed 100 km into the Grenville Orogen and these strata can be traced discontinuously across the entire width of the orogen (see Stockwell 1969). The geological fit is so good along much of the Front, and characteristic rock types can be traced for such great distances southward, that an explanation of 'anomalous' Grenville paleomagnetic pole positions by movement of the orogen several thousand kilometres (Irving *et al.* 1972) is exceedingly unlikely. An explanation of the 'anomalous' poles must be sought elsewhere.

The mechanism suggested by Dewey and Burke (1973) comes closest to explaining various puzzling aspects of Grenville tectonic history. These authors have suggested that the Grenville is comparable to the Tibetan Plateau. If this is the case, the character of the Grenville resulted from continental collision. The Grenville Front is not, however, a collision suture, but is analogous to the northern margin of the Tibetan Plateau; so the collision suture would lie somewhere southeast of the Grenville Orogen. As an ancient analog of the Tibetan Plateau, the Grenville, at an early stage would consist of tremendously thickened crust made up largely of reactivated Archean and Aphebian basement. Gradual erosion of this thickened crust would expose deeper zones of remobilized basement; and radiogenic ages obtained on rocks at the present surface would reflect resetting of the ages during a long process of uplift and cooling.

Paleomagnetic data are consistent with the Dewey and Burke (1973) hypothesis, if it is assumed firstly that the paleomagnetic pole for

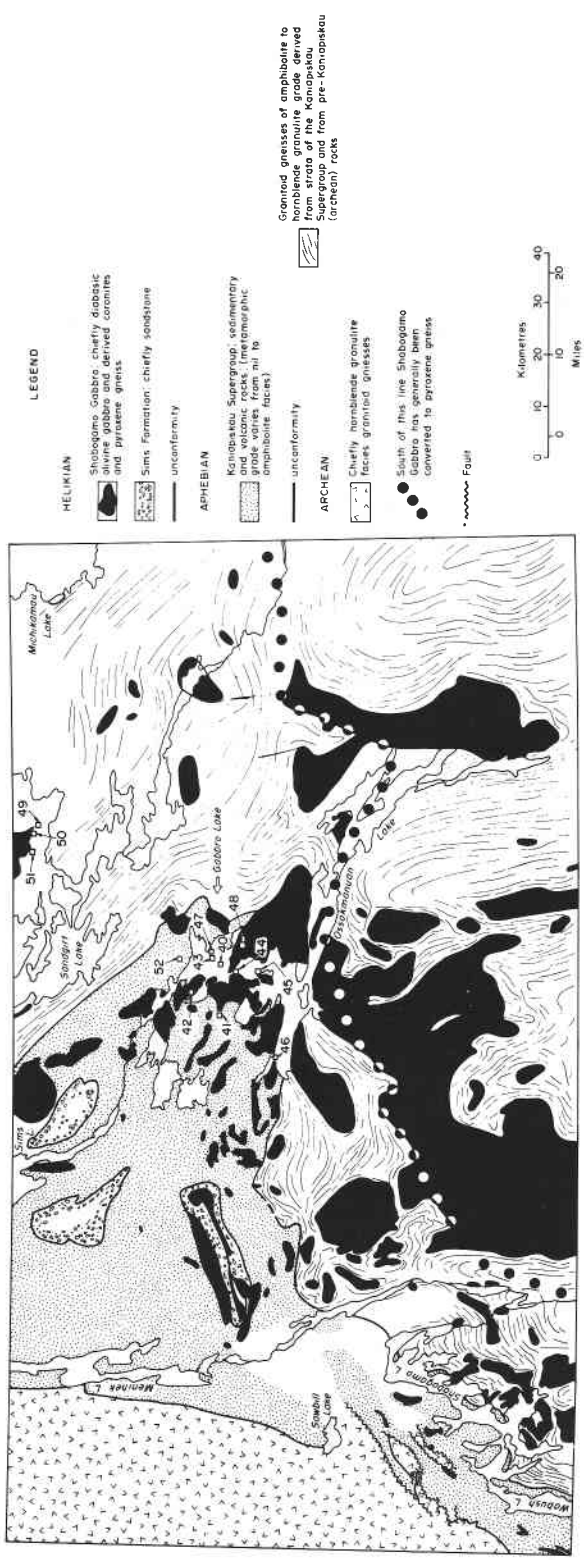


FIG. 6. The distribution of the Shabogamo Gabbro and sampling sites. Configuration of lakes in the Gabbro Lake region is quite different from the one shown here because of flooding resulting from the Churchill Falls project.

North America made an excursion deep into the South Pacific during the period 1250–1500 m.y. ago (Roy and Fahrig 1973), and secondly, the Grenville Orogen rotated clockwise 40–50° about a local axis, presumably during the continental collision that built the Grenville 'Plateau'.

The relationship between the various rotated Grenville Poles and the Grenville Loop (slightly modified from Roy and Fahrig 1973) is shown in Fig. 4. Notice that the development of the east component of the Mealy Mountain rocks appears to postdate the rotation. This suggests that the rocks around the perimeter of the anorthosite were intruded after the Grenville rotation or they were more susceptible than the anorthosite core to thermal events related to the building of the Grenville Plateau.

The northwest component of the anorthosite apparently 'looks through' these events to the time of original anorthosite crystallization.

Figure 8 is an attempt to diagramatically illustrate the geological relationships that existed prior to, and subsequent to, the continental collision (and related rotation) that may have initiated the last major tectonic events in the development of the Grenville Orogen. If the hypothesis is correct, the postulated thickening of continental crust should have been greater to the southwest because of the rotation, and should have also increased away (southeast) from the Front. In this case, the rocks of these areas should in general ex-

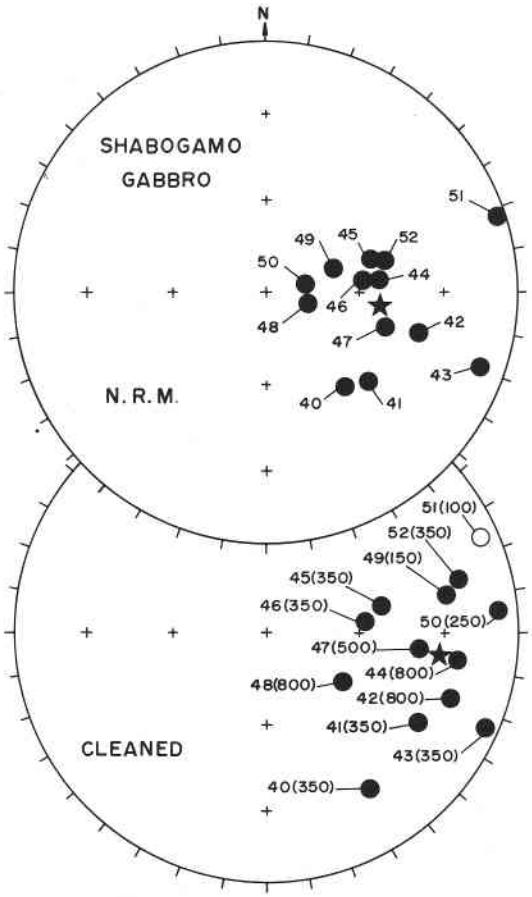


FIG. 7. Shabogamo Gabbro site mean directions before and after a.f. cleaning. Conventions as in Figs. 2 and 3.

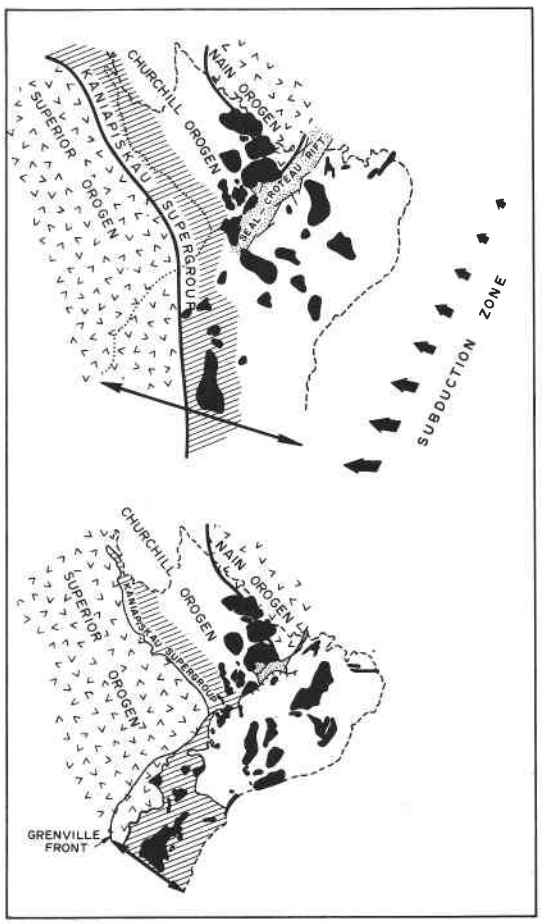


FIG. 8. The relationship of the Grenville and more ancient orogens prior to, and after the continental collision that caused the rotation of the Grenville Orogen and the building of the Grenville 'Plateau'.

TABLE 2. Directions of remanent magnetization before, (NRM), and after, magnetic cleaning

				Shabogamo Gabbro								
				NRM								
Site	N	n	D	I	$\delta°$	k	a.f.	D	I	$\delta°$	k	δ_R^S
40	6	6	139	49	12	42	350	145	26	11	59	4.3
41	7	5	140	45	51	2	350	120	30	9.0	82	4.0
42	5	5	105	37	3	822	800	109	23	3.2	643	1.4
43	6	6	109	10	13	39	350	112	5	7.4	119	3.0
44	6	5	83	52	21	15	800	98	24	15	30	6.6
45	7	7	73	53	8	100	350	77	51	6.0	184	2.3
46	6	6	83	57	5	267	350	83	57	5.1	249	2.1
47	7	7	106	49	5	223	500	96	38	5.6	208	2.1
48	5	5	104	75	10	67	800	122	61	8.5	91	3.8
49	7	3	73	66	18	20	150	78	28	26	10	15
50	7	5	81	77	8	112	250	85	7	16	27	6.9
51	7	4	73	0	35	5	100	66	− 7	25	10	13
52	6	6	74	50	5	281	350	75	21	5.4	227	2.2
All cores 70			97	51	29	8		98	31	28	8	3.1
All sites 13			97	51	27	9		97	30	28	8	7.7
Mean (south) pole position from 13 cleaned sites listed above—								171.2 °W	10.1 °S	$k = 10$		$\alpha°_{95} = 12$

NOTE: Symbols are the same as those used in Table 1.

hibit lower K-Ar ages, because present erosion surfaces in the southwest and away from the Front would have passed through the critical temperature for argon retention later than would those of the northeast Grenville. Evidence of late tectonic intrusions should be more abundant toward the southwest and away from the Front and older rocks, such as the anorthosites should show greater evidence of deformation and metamorphism and be less likely to retain the original T.R.M. components.

The age gradient (thermochron) map of the Grenville Orogen (Harper 1967), although based on inadequate data, appears to support this hypothesis, as does the unaltered nature of the Mealy anorthosite relative to anorthosite to the southwest. Furthermore, the Mealy anorthosite appears to retain a primary T.R.M., which is lacking in the Lac St. Jean body to the southwest (unpublished data). The paleomagnetism of Grenville rocks appears to be systematically updated southwest and away from the Front much as the K-Ar ages are updated.

The above synthesis throws into doubt the existence of a 'Grenville Orogeny', if this is defined as a thermal and tectonic event with a peak at about 1000 m.y. The development of the Grenville 'Plateau' was probably completed 1350 m.y. ago and most radiogenic ages and paleomagnetic data reflect the gradual uplift and erosion of this thickened layer of continental crust.

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